Essber Ruiz Icochea

Data Scientist | AI Automation & Advanced Dashboards

robertoruizicochea@gmail.com | +51 904 850 299 | Huánuco, Peru · Remote | Upwork profile

Profile

Data scientist (MSc, University of Freiburg) with **8+ years** turning data into working tools: machine-learning models, **advanced custom dashboards that go beyond Power BI** (live data from multiple APIs, embedded ML predictions, interactive maps) and **AI-agent automations**. **Top Rated on Upwork** with **$100K+ earned**, 43 jobs, 1,700+ hours and a **5.0 rating** across 29 client reviews. Former data scientist in compliance analytics at **BCP**, Peru's largest bank.

Core Services & Expertise

AI Automation	Building and specializing AI agents, prompt engineering, API keys and multi-API data integration, automated workflows
Advanced Dashboards	Custom interactive dashboards with live API data, built-in ML predictions and geospatial maps (R Shiny, Leaflet, D3.js, ggplot2)
ML & Statistics	XGBoost, deep learning (TensorFlow), time-series forecasting, failure prediction, fraud detection, segmentation, network analysis, text mining
Data Engineering	ETL pipelines, AWS (EC2), Linux servers (Ubuntu, CentOS)
Domain Knowledge	Finance & trading (strategy backtesting); crypto & the Bitcoin ecosystem (Ordinals, BRC-20, Runes); environmental & forest data

Experience

Freelance Data Scientist & AI Automation Consultant • Upwork (Remote) 08/2021 – Present

Top Rated since Dec 2021 · $100K+ earned · 43 jobs · 1,700+ hours · 5.0 rating (29 reviews)

- Engineered a **solar-site analysis tool on Amazon EC2** for a US client on the public NSRDB solar database (big data, R, web UI).
- Modeled **catastrophic failures in diesel engines** with ML (Python, TensorFlow, R).
- Deployed **ETL pipelines and machine-learning models on AWS** for an Upwork enterprise client.
- Designed **interactive map-and-reports web apps** with Leaflet and HTML popups; migrated Tableau work to R.
- Backtested **trading strategies**; developed multi-series **time-series forecasts**; delivered credit/loan analytics for a lending startup.

"Perhaps the best contractor I have ever hired in my many years of using Upwork." – US client

Data Scientist – Compliance Analytics (Deputy Manager) • BCP – Banco de Crédito del Perú, Lima 08/2018 – 03/2021

- Built supervised and unsupervised ML models for financial compliance; advanced visualizations for management.

Information Processing Specialist • Gestión Pesquera Sostenible, Lima 03/2018 – 07/2020

- Modeled Pacific Ocean temperature/salinity profiles; automated forecasts and real-time monitoring of oceanographic phenomena.

Earlier Career – Forestry & Research • Peru, Ecuador, Germany, Panama 2008 – 2013

Stable-isotope & DNA research to certify wood origin (Frankfurt/Lima); forest management at Forest Finance GmbH & Cambugan Foundation.

Technical Skills & Languages

Technical: R (expert), Python, AWS, TensorFlow, XGBoost, R Shiny, Leaflet, D3.js, ggplot2, LaTeX, Linux, SQL (basic), Spark (basic)

Languages: Spanish (native) · English (fluent/professional) · German (fluent/professional)

Education & Certifications

Master of Science (MSc) in Forest Ecology and Management • University of Freiburg, Germany

Forestry Engineer • Universidad Nacional Agraria La Molina, Peru

Certifications: IBM Deep Learning (2017), Big Data Foundations (2016), Data Science Foundations (2016) · SAP Business Intelligence, Alfa Training (2017)

Also manages 80 ha of natural forest in the Peruvian Amazon.